# SVR

*class* sklearn.svm.**SVR**(*, *kernel*='rbf', *degree*=3, *gamma*='scale', *coef0*=0.0, *tol*=0.001, *C*=1.0, *epsilon*=0.1, *shrinking*=True, *cache_size*=200, *verbose*=False, *max_iter*=-1) [source]

Epsilon-Support Vector Regression.

The free parameters in the model are C and epsilon.

The implementation is based on libsvm. The fit time complexity is more than quadratic with the number of samples which makes it hard to scale to datasets with more than a couple of 10000 samples. For large datasets consider using `LinearSVR` or `SGDRegressor` instead, possibly after a `Nystroem` transformer or other Kernel Approximation.

Read more in the User Guide.

**Parameters:**

**kernel** : *{'linear', 'poly', 'rbf', 'sigmoid', 'precomputed'} or callable, default='rbf'*

Specifies the kernel type to be used in the algorithm. If none is given, 'rbf' will be used. If a callable is given it is used to precompute the kernel matrix. For an intuitive visualization of different kernel types see Support Vector Regression (SVR) using linear and non-linear kernels

**degree** : *int, default=3*

Degree of the polynomial kernel function ('poly'). Must be non-negative. Ignored by all other kernels.

**gamma** : *{'scale', 'auto'} or float, default='scale'*

Kernel coefficient for 'rbf', 'poly' and 'sigmoid'.

- if `gamma='scale'` (default) is passed then it uses 1 / (n_features * X.var()) as value of gamma,
- if 'auto', uses 1 / n_features
- if float, must be non-negative.

> ! ***Changed in version 0.22:*** The default value of `gamma` changed from 'auto' to 'scale'.

**coef0** : *float, default=0.0*

Independent term in kernel function. It is only significant in 'poly' and 'sigmoid'.

**tol** : *float, default=1e-3*

Tolerance for stopping criterion.

**C** : *float, default=1.0*

Regularization parameter. The strength of the regularization is inversely proportional to C. Must be strictly positive. The penalty is a squared l2. For an intuitive visualization of the effects of scaling the regularization parameter C, see Scaling the regularization parameter for SVCs.

**epsilon** : *float, default=0.1*

Epsilon in the epsilon-SVR model. It specifies the epsilon-tube within which no penalty is associated in the training loss function with points predicted within a distance epsilon from the actual value. Must be non-negative.

**shrinking** : *bool, default=True*

Whether to use the shrinking heuristic. See the User Guide.

**cache_size** : *float, default=200*

Specify the size of the kernel cache (in MB).

**verbose** : *bool, default=False*

Enable verbose output. Note that this setting takes advantage of a per-process runtime setting in libsvm that, if enabled, may not work properly in a multithreaded context.

**max_iter** : *int, default=-1*

Hard limit on iterations within solver, or -1 for no limit.

**Attributes:**

`coef_` : *ndarray of shape (1, n_features)*

Weights assigned to the features when `kernel="linear"`.

**dual_coef_** : *ndarray of shape (1, n_SV)*

Coefficients of the support vector in the decision function.

**fit_status_** : *int*

0 if correctly fitted, 1 otherwise (will raise warning)

**intercept_** : *ndarray of shape (1,)*

Constants in decision function.

**n_features_in_** : *int*

Number of features seen during fit.

> *Added in version 0.24.*

**feature_names_in_** : *ndarray of shape (* `n_features_in_` *,)*

Names of features seen during fit. Defined only when `X` has feature names that are all strings.

> *Added in version 1.0.*

**n_iter_** : *int*

Number of iterations run by the optimization routine to fit the model.

> *Added in version 1.1.*

`n_support_` : *ndarray of shape (1,), dtype=int32*

Number of support vectors for each class.

**shape_fit_** : *tuple of int of shape (n_dimensions_of_X,)*

Array dimensions of training vector `X`.

**support_** : *ndarray of shape (n_SV,)*

Indices of support vectors.

**support_vectors_** : *ndarray of shape (n_SV, n_features)*

Support vectors.

See also

`NuSVR`

Support Vector Machine for regression implemented using libsvm using a parameter to control the number of support vectors.

`LinearSVR`

Scalable Linear Support Vector Machine for regression implemented using liblinear.

## References

[1] LIBSVM: A Library for Support Vector Machines

[2] Platt, John (1999). "Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods"

## Examples

```
>>> from sklearn.svm import SVR
>>> from sklearn.pipeline import make_pipeline
>>> from sklearn.preprocessing import StandardScaler
>>> import numpy as np
>>> n_samples, n_features = 10, 5
>>> rng = np.random.RandomState(0)
>>> y = rng.randn(n_samples)
>>> X = rng.randn(n_samples, n_features)
>>> regr = make_pipeline(StandardScaler(), SVR(C=1.0, epsilon=0.2))
>>> regr.fit(X, y)
Pipeline(steps=[('standardscaler', StandardScaler()),
                ('svr', SVR(epsilon=0.2))])
```

### fit(*X, y, sample_weight=None*) [source]

Fit the SVM model according to the given training data.

**Parameters:**

**X** : ***{array-like, sparse matrix} of shape (n_samples, n_features) or (n_samples, n_samples)***

Training vectors, where `n_samples` is the number of samples and `n_features` is the number of features. For kernel="precomputed", the expected shape of X is (n_samples, n_samples).

**y** : ***array-like of shape (n_samples,)***

Target values (class labels in classification, real numbers in regression).

**sample_weight** : ***array-like of shape (n_samples,), default=None***

Per-sample weights. Rescale C per sample. Higher weights force the classifier to put more emphasis on these points.

**Returns:**

**self : *object***

Fitted estimator.

### Notes

If X and y are not C-ordered and contiguous arrays of np.float64 and X is not a scipy.sparse.csr_matrix, X and/or y may be copied.

If X is a dense array, then the other methods will not support sparse matrices as input.

## `get_metadata_routing()` [source]

Get metadata routing of this object.

Please check User Guide on how the routing mechanism works.

**Returns:**

**routing : *MetadataRequest***

A `MetadataRequest` encapsulating routing information.

## `get_params(deep=True)` [source]

Get parameters for this estimator.

**Parameters:**

**deep : *bool, default=True***

If True, will return the parameters for this estimator and contained subobjects that are estimators.

**Returns:**

**params : *dict***

Parameter names mapped to their values.

## `predict(X)` [source]

Perform regression on samples in X.

For an one-class model, +1 (inlier) or -1 (outlier) is returned.

**Parameters:**

**X : *{array-like, sparse matrix} of shape (n_samples, n_features)***

For kernel="precomputed", the expected shape of X is (n_samples_test, n_samples_train).

**Returns:**

**y_pred** : *ndarray of shape (n_samples,)*

The predicted values.

`score(X, y, sample_weight=None)` [source]

Return coefficient of determination on test data.

The coefficient of determination, $R^2$, is defined as $(1 - \frac{u}{v})$, where $u$ is the residual sum of squares `((y_true - y_pred)** 2).sum()` and $v$ is the total sum of squares `((y_true - y_true.mean()) ** 2).sum()`. The best possible score is 1.0 and it can be negative (because the model can be arbitrarily worse). A constant model that always predicts the expected value of `y`, disregarding the input features, would get a $R^2$ score of 0.0.

**Parameters:**

**X** : *array-like of shape (n_samples, n_features)*

Test samples. For some estimators this may be a precomputed kernel matrix or a list of generic objects instead with shape `(n_samples, n_samples_fitted)`, where `n_samples_fitted` is the number of samples used in the fitting for the estimator.

**y** : *array-like of shape (n_samples,) or (n_samples, n_outputs)*

True values for `X`.

**sample_weight** : *array-like of shape (n_samples,), default=None*

Sample weights.

**Returns:**

**score** : *float*

$R^2$ of `self.predict(X)` w.r.t. `y`.

## Notes

The $R^2$ score used when calling `score` on a regressor uses `multioutput='uniform_average'` from version 0.23 to keep consistent with default value of `r2_score`. This influences the `score` method of all the multioutput regressors (except for

`MultiOutputRegressor` ).

**set_fit_request**(*, *sample_weight: bool | None | str = '$UNCHANGED$'*) → SVR [source]

Configure whether metadata should be requested to be passed to the `fit` method.

Note that this method is only relevant when this estimator is used as a sub-estimator within a meta-estimator and metadata routing is enabled with `enable_metadata_routing=True` (see `sklearn.set_config` ). Please check the User Guide on how the routing mechanism works.

The options for each parameter are:

- `True` : metadata is requested, and passed to `fit` if provided. The request is ignored if metadata is not provided.
- `False` : metadata is not requested and the meta-estimator will not pass it to `fit` .
- `None` : metadata is not requested, and the meta-estimator will raise an error if the user provides it.
- `str` : metadata should be passed to the meta-estimator with this given alias instead of the original name.

The default ( `sklearn.utils.metadata_routing.UNCHANGED` ) retains the existing request. This allows you to change the request for some parameters and not others.

> ! *Added in version 1.3.*

**Parameters:**

**sample_weight** : *str, True, False, or None, default=sklearn.utils.metadata_routing.UNCHANGED*

Metadata routing for `sample_weight` parameter in `fit` .

**Returns:**

**self** : *object*

The updated object.

**set_params**(****params*) [source]

Set the parameters of this estimator.

The method works on simple estimators as well as on nested objects (such as `Pipeline`). The latter have parameters of the form `<component>__<parameter>` so that it's possible to update each component of a nested object.

**Parameters:**

****params : *dict***

Estimator parameters.

**Returns:**

**self : *estimator instance***

Estimator instance.

`set_score_request(*, sample_weight: bool | None | str = '$UNCHANGED$') → SVR` [source]

Configure whether metadata should be requested to be passed to the `score` method.

Note that this method is only relevant when this estimator is used as a sub-estimator within a meta-estimator and metadata routing is enabled with `enable_metadata_routing=True` (see `sklearn.set_config`). Please check the User Guide on how the routing mechanism works.

The options for each parameter are:

- `True` : metadata is requested, and passed to `score` if provided. The request is ignored if metadata is not provided.
- `False` : metadata is not requested and the meta-estimator will not pass it to `score`.
- `None` : metadata is not requested, and the meta-estimator will raise an error if the user provides it.
- `str` : metadata should be passed to the meta-estimator with this given alias instead of the original name.

The default ( `sklearn.utils.metadata_routing.UNCHANGED` ) retains the existing request. This allows you to change the request for some parameters and not others.

> ***Added in version 1.3.***

**Parameters:**

**sample_weight** : ***str, True, False, or None, default=sklearn.utils.metadata_routing.UNCHANGED***

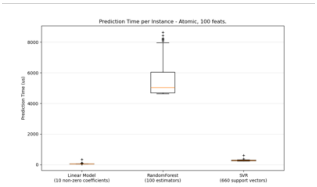
Metadata routing for `sample_weight` parameter in `score`.

**Returns:**

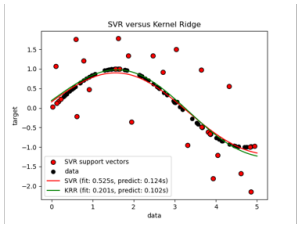
**self** : ***object***

The updated object.

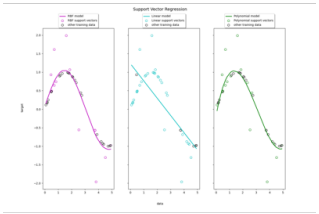
# Gallery examples




Prediction Latency




Comparison of kernel ridge regression and SVR




Support Vector Regression (SVR) using linear and non-linear kernels

Previous
SVC

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l1_min_c

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